PARTIAL CIRCUIT MILESTONE — NOT A RELEASE DESIGN

Global nets join sheets. GEN_ and HUB_ segregate control and phase nets. DC_LINK / PGND shared.

36–58.4 V screening range only. No validated current rating. No PCB or routing.

Offboard mandatory: fuse, reverse protection, precharge, disconnect, OV chopper and independent inhibit.

GEN DRV8353S driver and three low-side current channels

File: gen-driver.kicad_sch

GEN 100 V MOSFET bridge — schematic only

File: gen-bridge.kicad_sch

HUB DRV8353S driver and three low-side current channels

File: hub-driver.kicad_sch

HUB 100 V MOSFET bridge — schematic only

File: hub-bridge.kicad_sch

Unfinished safety / control / supply boundaries

File: external-boundaries.kicad_sch

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /
File: powerboard-reva.kicad_sch

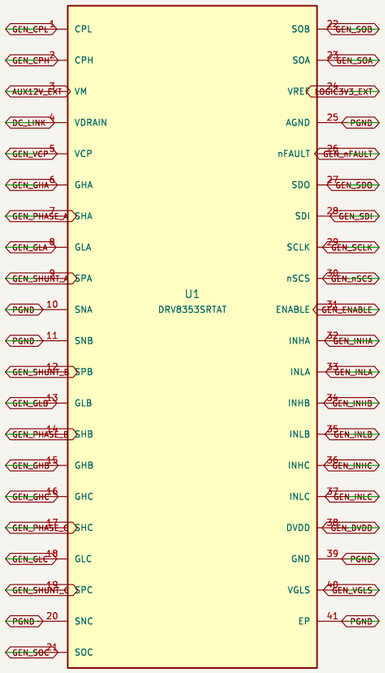
Title: Pedal-by-wire: hierarchical engineering review

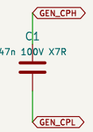
Size: A3	Date:	Rev: A-PARTIAL
KiCad E.D.A. 9.0.9		Id: 1/6

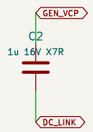
TI SLVSDY6A pp6–7 pin table. 40-pin RTA + exposed pad. SPI variant, NOT hardware variant.

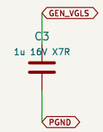
VM supplied by external protected 12 V. VREF=3V3 (CSA midpoint internally VREF/2).

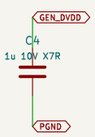
RELEASE BLOCKERS: no independent fault latch/watchdog/OV inhibit; no MCU pin allocation.

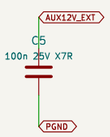


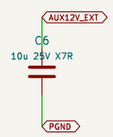


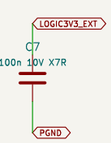




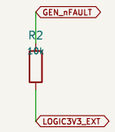


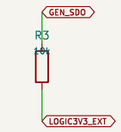


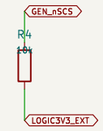












ENABLE pull-down defaults off; not a safety-rated inhibit. SPI configuration and CSA gain20 not yet validated.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

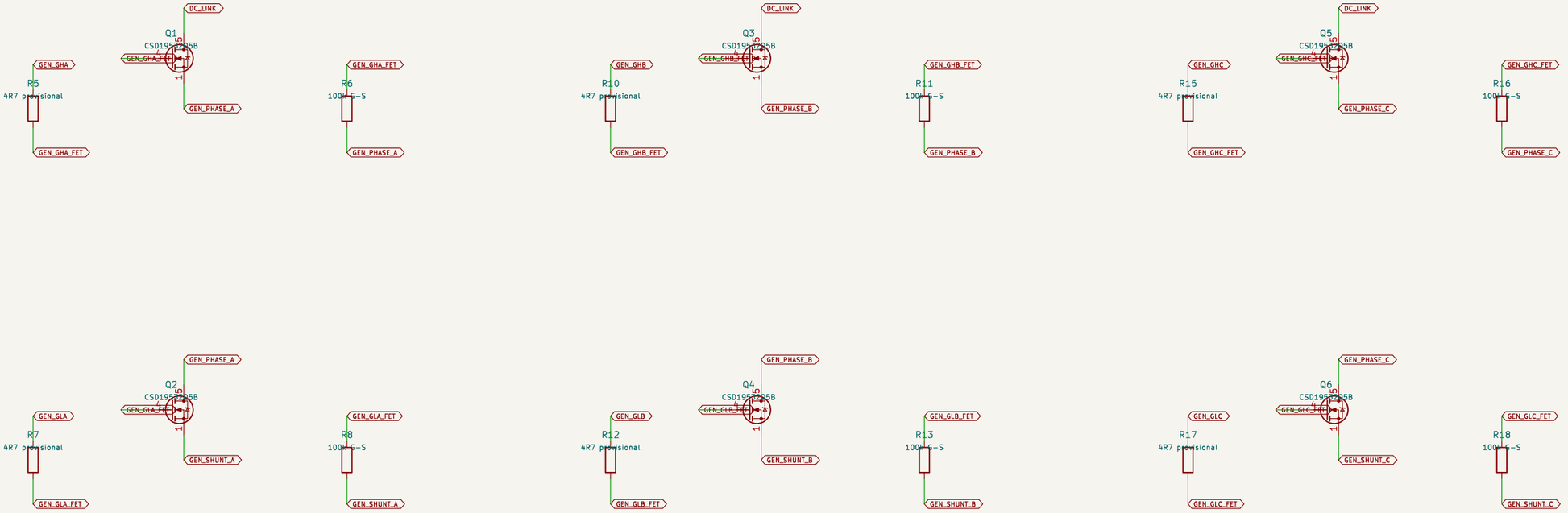
Sheet: /GEN DRV8353S driver and three low-side current channels/
File: gen-driver.kicad_sch

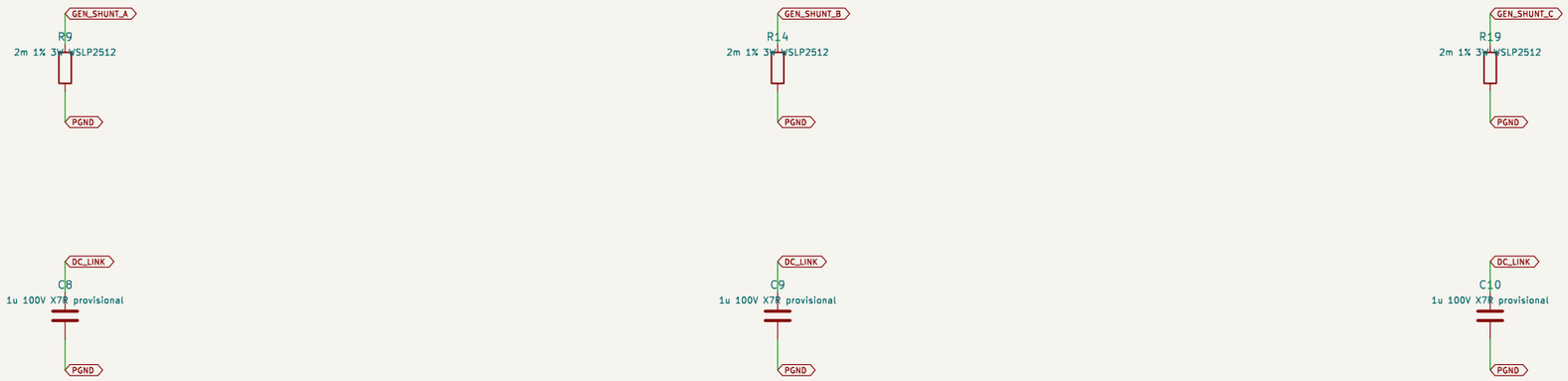
Title: GEN DRV8353S driver and three low-side current channels

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 2/6

CSD19532Q5B stock KiCad symbol / pad map. Gates: 4; drains: 5; sources: 1,2,3.

Labels are electrical connections. Rgate=4R7 provisional. Kelvin shunts: no routing qualified.





Shunt amplifier SPx taps SHUNT_x; SNx taps PGND: Kelvin layout mandatory. 15A RMS => 0.45W per shunt.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /GEN 100 V MOSFET bridge — schematic only/
File: gen-bridge.kicad_sch

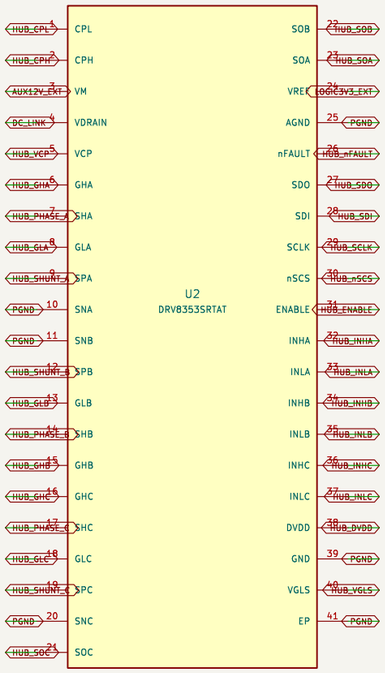
Title: GEN 100 V MOSFET bridge — schematic only

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 3/6

TI SLVSDY6A pp6–7 pin table. 40-pin RTA + exposed pad. SPI variant, NOT hardware variant.

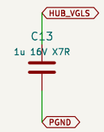
VM supplied by external protected 12 V. VREF=3V3 (CSA midpoint internally VREF/2).

RELEASE BLOCKERS: no independent fault latch/watchdog/OV inhibit; no MCU pin allocation.



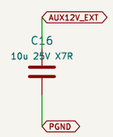


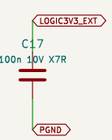




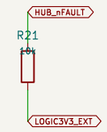


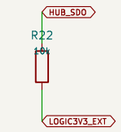


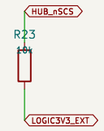












ENABLE pull-down defaults off; not a safety-rated inhibit. SPI configuration and CSA gain20 not yet validated.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

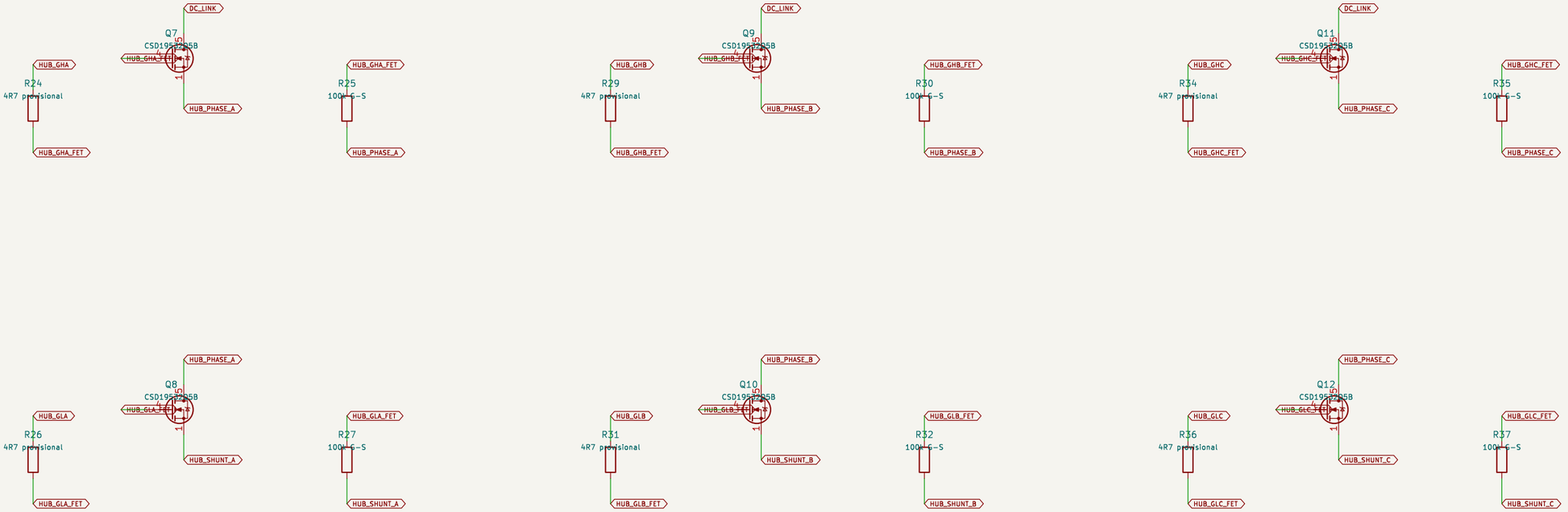
Sheet: /HUB DRV8353S driver and three low-side current channels/
File: hub-driver.kicad_sch

Title: HUB DRV8353S driver and three low-side current channels

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 4/6

CSD19532Q5B stock KiCad symbol / pad map. Gates: 4; drains: 5; sources: 1,2,3.

Labels are electrical connections. Rgate=4R7 provisional. Kelvin shunts: no routing qualified.



Shunt amplifier SPx taps SHUNT_x; SNx taps PGND: Kelvin layout mandatory. 15A RMS => 0.45W per shunt.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /HUB 100 V MOSFET bridge — schematic only/
File: hub-bridge.kicad_sch

Title: HUB 100 V MOSFET bridge — schematic only

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 5/6

ALL CONNECTORS BELOW ARE OFFBOARD INTERFACES, NOT IMPLEMENTED ICs OR SAFETY CIRCUITS.

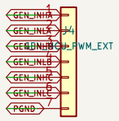
DO NOT ENERGIZE. Mandatory upstream safety, chopper, MCU and auxiliary circuitry not included.

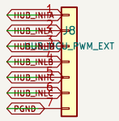




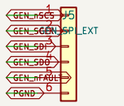


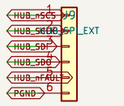


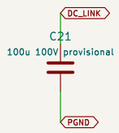


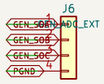


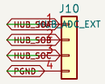












Missing: source fuse, reverse blocking, precharge, DC contactor, bus current sensor and discharge.

Missing: independent OV chopper/resistor, OC comparator/latch, brake interlock, watchdog and UVLO.

Missing: STM32 timer/ADC mux, Hall/encoder sensors, thermal sensors, storage/BMS charge acceptance.

PGND is common, not galvanically isolated. Supply sequencing and back-power paths unqualified.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

Sheet: /Unfinished safety / control / supply boundaries/
File: external-boundaries.kicad_sch

Title: Unfinished safety / control / supply boundaries

Size: A3	Date:	Rev: A-PARTIAL
KiCad E.D.A. 9.0.9		Id: 6/6